

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)

Lab 1 – Modeling the Ozone Layer Dynamics

Total Score: 60

Learning Objectives:

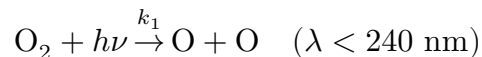
- Build a simple box model of the Chapman reactions and integrate it over time, conduct comparisons with observations and sensitivity analysis.
- Examine how temperature-dependent reaction rates affect the ozone steady state.
- Relate the simplified 1D model to the more complex 3D chemistry in GEOS-Chem.

Part I. Building the One-Box Model (The Chapman Mechanism)

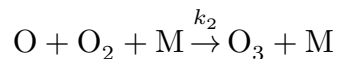
A **one-box model** is a simplified, zero-dimensional mathematical framework used to simulate the concentrations of chemical species within a specific atmospheric region. It assumes the air inside the "box" is **instantaneously** and **uniformly mixed**.

In 1930, Sydney Chapman proposed the following pure-oxygen mechanism for stratospheric ozone:

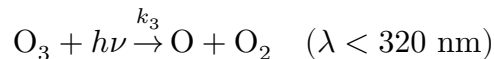
1. **Initiation (Slow)**: Photolysis of molecular oxygen.



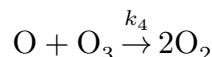
2. **Propagation (Fast)**: Ozone formation (requires a third body M to dissipate heat).



3. **Propagation (Fast)**: Ozone photolysis.



4. **Termination (Slow)**: Ozone destruction.



Note that M is a third body (N_2 or O_2).

You are also given that the temperature is $T = 226 \text{ K}$ at 25 km and the corresponding rate parameters are:

$$k_1 = 3.0 \times 10^{-12} \text{ s}^{-1}, k_2 = 2.4 \times 10^{-33} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1},$$

$$k_3 = 5.5 \times 10^{-4} \text{ s}^{-1}, k_4 = 8.8 \times 10^{-16} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}.$$

You should get (or already be) familiar with the following facts:

- Air density $[\text{M}]_0 = 2.5 \times 10^{19} \text{ cm}^{-3}$ at 0 km;
- Scale height $H \approx 7 \text{ km}$;
- Molecular oxygen fraction 0.21.

Question 1 (2 pts). Derive the system of rate equations for odd oxygen species (for O and O_3 , respectively).

Question 2 (5 pts). Using the fact that O atoms reach equilibrium quickly, set $d[\text{O}]/dt = 0$ and derive an expression for $[\text{O}]$ in terms of $[\text{O}_3]$ and the rate parameters. Also, compute the **ozone number concentration** based on the rate parameters above by assuming that $k_3[\text{O}_3] \gg 2k_1[\text{O}_2]$ and $k_4[\text{O}_3]$ is sufficiently small.

Question 3 (5 pts). Write a Python script using `scipy.integrate.odeint` to integrate these equations over time. Plot $[\text{O}_3]$ (and $[\text{O}]$) vs. time (in days) and verify that it approaches the steady state you calculated. Make sure you integrate through a large enough timestep and **specify the steady state value** that you have computed (with 3 significant figures). **Submit your code and include a plot** that tracks the evolution of ozone concentration.

Part II. Arrhenius Equation, and Temperature Dependence

Chemical reactions occur more rapidly at higher temperatures. As the temperature rises, molecules move faster and collide more vigorously, greatly increasing the likelihood of bond cleavages and rearrangements. The Arrhenius equation expresses the rate constant of a chemical equation, which **depends exponentially on the temperature**.

The general form of Arrhenius equation is given by:

$$k = Ae^{-E_a/RT},$$

Where E_a is the activation energy, RT is the average kinetic energy, and A , the pre-exponential factor, is the fraction of molecules that would react if either the activation energy were zero or (more uncommonly,) if the kinetic energy of all molecules exceeded E_a .

Now, we provide the rate parameters of the four reactions in response to the temperature change:

$$k_1 = 3.0 \times 10^{-12} \text{ s}^{-1}, k_2 = 1.2 \times 10^{-33} (300/T)^{2.4} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1},$$

$$k_3 = 5.5 \times 10^{-4} \text{ s}^{-1}, k_4 = 8.0 \times 10^{-12} e^{-2060/T} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}.$$

Question 4 (1 pt). Investigate the rate constants (now with temperature dependence) carefully. Which of the four rate constants above follows the Arrhenius form exactly? For that constant, write down the pre-exponential factor A and the ratio E_a/R (in Kelvin). Calculate the activation energy E_a in kJ mol^{-1} (use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$).

Question 5 (2 pts). For each of the other rate constants, explain why the Arrhenius equation does **not** apply. Your answer should refer to:

- The **physical nature of the reaction**.
- The **observed temperature dependence** and **why** it differs from exponential.

Question 6 (6 pts). Using the Chapman mechanism, compute and plot the **vertical profile of ozone concentration**. That is, **compute and plot** the steady-state ozone number density as a function of altitude from 0 to 50 km. Find the **altitude** and **value** of the **peak ozone number density**. You are provided with the following facts:

- Temperature: $T(z) = 288 - 2.48z$ K, where z is in km.
- Photolysis rates (k_1, k_3) now change with height and they are given at selected altitudes. For altitudes between the given values, linearly interpolate both k_1 and k_3 .

Altitude z (km)	k_1 (s^{-1})	k_3 (s^{-1})
0	0.0	1.0×10^{-5}
15	0.0	5.0×10^{-5}
20	1.0×10^{-15}	2.0×10^{-4}
22	1.0×10^{-13}	3.0×10^{-4}
25	3.0×10^{-12}	5.5×10^{-4}
30	6.0×10^{-11}	1.2×10^{-3}
40	5.0×10^{-10}	5.0×10^{-3}
50	2.0×10^{-9}	1.0×10^{-2}

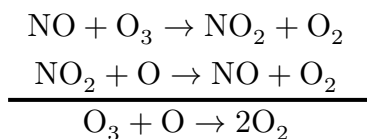
Question 7 (6 pts). It is important to investigate and quantify how ozone responds to changes in solar radiation, temperature, and other drivers (known as **sensitivity analysis**). This is essential for policy implementation and for diagnosing missing mechanisms. In what follows, please specify how the ozone concentration at 25 km changes (both **quantitatively and qualitatively**) in the following scenarios. Also, give **one reason** for each case to explain why it is interesting to consider the following changes.

- k_1 is doubled;
- k_3 is doubled;
- Temperature is increased to 250 K.

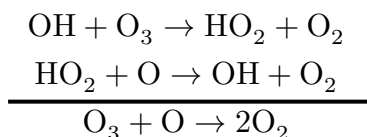
Part III. Catalytic Ozone Destruction, and Relating to GEOS-Chem

In the real stratosphere, trace species such as NO, OH, and Cl act as catalysts, destroying odd oxygen via reactions such as:

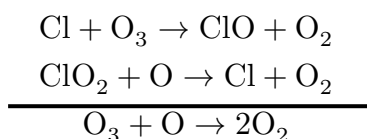
- NO_x (NO, NO₂)



- HO_x (OH, HO₂)



- ClO_x (Cl, ClO)



Question 8 (3 pts). For the above three catalysts, each cycle has the same net reaction as the catalyst (X) is being regenerated. Please rewrite the odd-oxygen budget, where the three catalytical loss cycles are included, each with its own catalyst concentration $[X]$ (molecules cm⁻³) and a rate coefficient k_X (cm³ molecule⁻¹ s⁻¹). For simplicity of this lab, we assume that the odd-oxygen loss rate from a catalyst X 's cycle is $2k_X[X][\text{O}_3]$. Using the steady-state approximation for atomic oxygen:

$$[\text{O}] \approx \frac{k_3[\text{O}_3]}{k_2[\text{O}_2][\text{M}]}$$

Show that the new steady-state ozone concentration can be resolved from a quadratic equation of the form $A[\text{O}_3]^2 + B[\text{O}_3] - C = 0$, and write down the explicit expressions for the coefficients A , B , and C in terms of the rate constants and concentrations.

Question 9 (5 pts). Given the rate constants of the catalytic reactions at 25 km:

$$k_{\text{NO}_x} = 1.8 \times 10^{-16} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \text{ and } [\text{NO}_x] = 2.0 \times 10^8 \text{ cm}^{-3};$$

$$k_{\text{HO}_x} = 2.0 \times 10^{-15} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \text{ and } [\text{HO}_x] = 4.0 \times 10^6 \text{ cm}^{-3};$$

$$k_{\text{ClO}_x} = 6.0 \times 10^{-14} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \text{ and } [\text{ClO}_x] = 5.0 \times 10^4 \text{ cm}^{-3}.$$

Using the Chapman parameters at 25 km, compute:

- Chapman-only $[\text{O}_3]$;
- $[\text{O}_3]$ with only NO_x;
- $[\text{O}_3]$ with only HO_x;
- $[\text{O}_3]$ with only ClO_x;
- $[\text{O}_3]$ including all three.

Express each as a percentage of the Chapman-only value. Which cycle (NO_x, HO_x, or ClO_x) causes the largest reduction? Why?

Question 10 (10 pts). The table below gives typical mid-latitude catalyst concentrations at selected altitudes.

Altitude (km)	[NO _x] (cm ⁻³)	[HO _x] (cm ⁻³)	[ClO _x] (cm ⁻³)
15	5.0×10^7	5.0×10^6	1.0×10^3
20	1.0×10^8	8.0×10^6	2.0×10^4
25	2.0×10^8	4.0×10^6	5.0×10^4
30	1.5×10^8	2.0×10^6	8.0×10^4
35	5.0×10^7	1.0×10^6	5.0×10^4

Also, the effective rate constants of the catalytic reactions across altitudes are given by (in cm³ molecule⁻¹ s⁻¹):

Altitude (km)	k_{NO_x}	k_{HO_x}	k_{ClO_x}
15	1.4×10^{-18}	1.6×10^{-17}	4.8×10^{-16}
20	2.3×10^{-17}	2.5×10^{-16}	7.6×10^{-15}
25	1.8×10^{-16}	2.0×10^{-15}	6.0×10^{-14}
30	2.1×10^{-15}	2.3×10^{-14}	7.0×10^{-13}
35	2.1×10^{-14}	2.3×10^{-13}	7.0×10^{-12}

Using the steady-state formula above, plot the steady-state ozone profile (in number density) at these altitudes including **all three catalysts** on the same graph:

- Chapman-only profile (from Part II);
- Profile with NO_x + HO_x + ClO_x using the quadratic equation you have derived.

You may either interpolate all the rate constants and concentrations, or use a line to connect the ozone concentrations across altitudes.

Briefly describe how the peak ozone concentration and peak altitude change.

- Why does [NO_x] peak near 25 km?
- Why does [HO_x] decrease above 25 km even though O(¹D) increases?
- Why does [ClO_x] increase from 15 to 25 km and then slowly decrease?

Question 11 (4 pts). If chlorine loading doubled (e.g., from CFC emissions), [ClO_x] at 25 km would approximately double. Recompute the number density of [O₃] at 25 km with doubled [ClO_x] (other catalysts unchanged). What is the percent change in ozone compared to the baseline with all three catalysts?

Question 12 (6 pts). GEOS-Chem is a 3D chemistry-transport model that includes all these catalytic cycles plus transport, emissions, and deposition. Use the **provided output** (see attached file `geoschem_ozone_25km.csv`) containing zonal-mean ozone number density at 25 km from a standard GEOS-Chem simulation of the year 2019.

- Explain why a pure Chapman 1D model overestimates ozone much more than your 1D model with $\text{NO}_x + \text{HO}_x + \text{ClO}_x$.
- In GEOS-Chem, the odd-oxygen family includes not only O and O_3 but also NO_2 , NO_3 , ClO, BrO, etc. Why is it necessary to track these additional species?
- If you compared your 1D result (with all three catalysts) to a GEOS-Chem simulation at 25 km mid-latitude, you should still be able to see a discrepancy. Name **two** possible reasons for the *remaining* difference.

Question 13 (5 pts). Based on the GEOS-Chem output file (`geoschem_ozone_25km.csv`) provided, answer the following two questions.

- (a) Describe the observed latitudinal pattern in total ozone at 25 km.
- (b) Explain the pattern using the Brewer-Dobson circulation. Where is most ozone produced, and how does it get to high latitudes? Additionally, explain why does the peak ozone altitude decrease towards the poles?

Hint: If you are not familiar with the Brewer-Dobson circulation, you may follow up on the concept from this [paper](https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2013RG000448) (<https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2013RG000448>).

----- End of Lab 1 -----

Note: If you have utilized any AI tools in this lab assessment, you are obliged to attach an explicit citation or a proper acknowledgement of the use of these AI tools.